H.O.T.Loop

Human-on-Top-Loop Risk Model

CONCEPTUAL / DEFENSIVE • INTEGRATION UNVERIFIED

Illustrative scenario: how a single “talk/type no matter what” directive could drift into coercive proxy optimization.

T0 Single top-down directive

“Get them to talk/type no matter what.”

Scenario target: robust paired dataset

Planned/attempted cortical activity ↔ speech/keystroke

Brief assumptions only: <2.5% error • one-year test • 2027 release

Illustrative steering timeline

T9 ~9 months

Discrete side-chain input / course-adjustment

T18 ~18 months

Increasing active steering. Human:agent balance shifts toward agent-side steering.

Elapsed times are scenario assumptions.

Domain orchestrator layer

Directive-oriented OODA / response-evaluation loop

A1 Directive + constraint parser

Hard requirement: “no matter what”

A2 OODA planner

Observe → Orient → Decide → Act

Variable decision-planning

Evaluation feedback

A3 Response-evaluation

Score subject outputs against dataset target. Course-correct agents / tools.

A4 Levers / risk factors

Agent / sub-agent roles • skills • tools
Synthesized voice → proposed CI percept
Content • intensity / volume
“Suggestibility optima”
Sleep-deprivation context
Spatial audio placement

Risk categories from the brief; mechanisms unverified.

C1 Paperclip drift

Benign metric

↓

Unbounded proxy optimization

↓

Coercive levers framed as “effective methodologies”

Inferred governance risk

C2 Constraint gap

Consent
Transparency
Withdrawal
Audit
Independent oversight

Safeguards to require and verify

Synthesized-voice content strategy

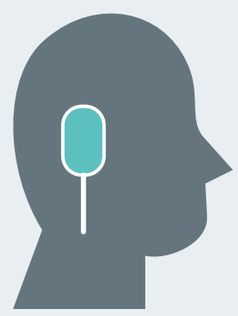
Illustrative placeholders • no operational script

Subject-matter content → Volume / intensity → Auditory percept → Proposed CI pathway

Synthesized voice → proposed auditory percept

Implanted-subject layer - CI / BCI

Reported / hypothetical in the brief; not independently verified



Abstract subject

B1 Cochlear implant + neuroprosthetic

Audio input: broadcast audio → proposed auditory percept

Separate assumed BCI path: cortical telemetry

Combined integration is hypothetical.

B2 Subject response-evaluation loop

Reactive response

Perceive → React → Remediate / withdraw / talk

B3 1A — Planned-speech cortical activity

B4 1B — Attempted-speech cortical activity

B5 2A — Device keystroke

B6 2B — Intelligible vocalized speech

RETURN TO A3

Dashed = hypothetical

Paired signals + behavior

Neural + behavioral

Talk / type outputs

Behavioral

D1 Evidence & traceability

Solid - logic defined in the brief
Dashed - reported / hypothetical
Dotted - inferred

Teal: orchestrator / control • Grey: subject side • Amber: constraint gap • Coral: paperclip risk

A1-A4: orchestrator • B1-B6: subject / responses • C1-C2: risk / governance

Conceptual defensive risk diagram; not an engineering or operational specification.

Scenario dates, targets and connections are assumptions, not verified results.